Laser-cut parts — 4 kg oval trebuchet

These files cut the trebuchet's two A-frame sides and its small parts from **four sheets of 6 mm plywood** on a **900 × 600 mm** bed. Each side is two layers glued together (12 mm). The arm stays a 40×20×1.6 mm aluminium tube, and the base is timber: the laser can't cut either at this size. Sizes follow pages 8–9 of the Build Guide, changed for the 4 kg limit: short arm 350, hang 280, and the axle 800 above the base rails.

What is in the pack

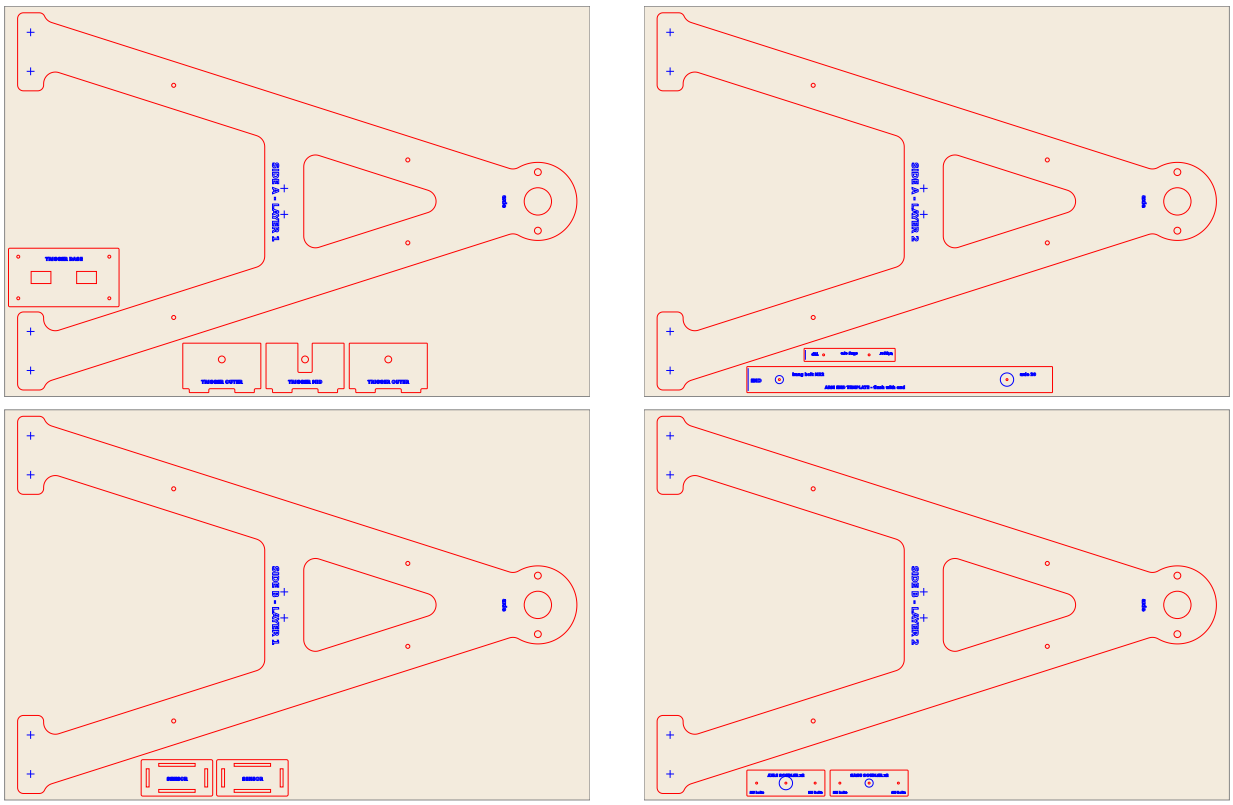
File (.dxf and .svg)	Material	Parts
00_Test_coupon_cut_first	6 mm ply offcut	Fit test: axle and pin holes, tab slots
01_Sheet_SIDE_A_LAYER_1	6 mm ply	A-frame layer; trigger block (3 layers) and trigger base
02_Sheet_SIDE_A_LAYER_2	6 mm ply	A-frame layer; arm drilling templates (end and tip)
03_Sheet_SIDE_B_LAYER_1	6 mm ply	A-frame layer; two sensor plates
04_Sheet_SIDE_B_LAYER_2	6 mm ply	A-frame layer; marking templates for the aluminium doubler plates
05_Fabric_sling_pouches	canvas or leather	Six sling pouches
gen.py	—	The script that makes the files. Every size is a setting at the top.

Colours: red = cut through (layer CUT); blue = engrave or score lightly (layer ENGRAVE: labels and the screw marks for the brackets). Run the blue layer first, then the red; inside cuts before outlines. Everything is in millimetres at 1 : 1 — check the import scale by measuring the 120 mm boss on screen.

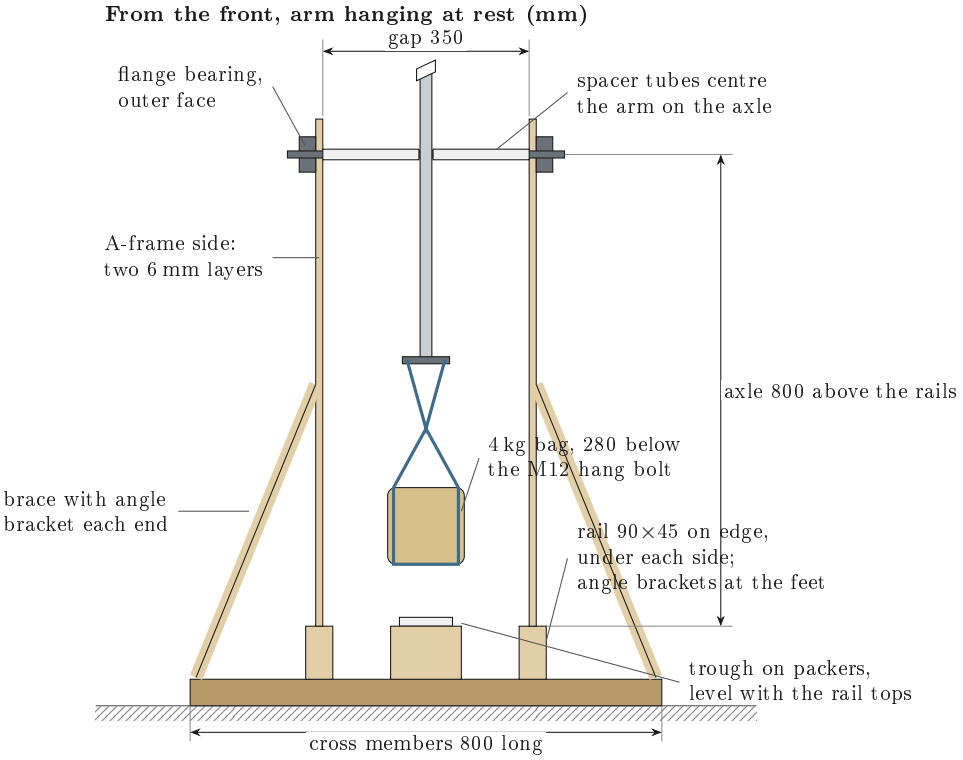
Before you cut the big sheets

1. **Cut the fit test** (00) from an offcut. The drawn hole sizes are nominal; your laser's kerf makes them a little bigger. Check which hole your 10 mm trigger pin slides through freely, and that the three trigger layers (glued, 18 mm) push into a tab slot snugly.
2. **Check your bearings.** The side panels are drawn for a 20 mm-bore **two-bolt flange bearing** with bolts **90 mm apart** (the common UCFL204 pattern) and a 42 mm clearance hole for its housing. Measure yours. If it differs, change `BEARING_BOLT_PITCH` and `BEARING_CLEAR` in `gen.py` and run it again.
3. If a fit is wrong, change the matching setting in `gen.py` (for example `PIN_D`, `T` for thicker ply) and run `python3 gen.py`. It re-nests the sheets and checks that no parts overlap or leave the sheet.

Sheet layouts



How it goes together



Assembly

1. **Glue each pair** of A-frame layers (A1+A2, B1+B2) with PVA. Push 6 mm dowels through the five dowel holes to line them up; engraved faces outward. Clamp flat on a bench overnight; trim the dowels flush.
2. **Base:** lay the three cross members (front, back, and one under the pivot line), then screw the two rails across them, on edge, 362 mm apart centre to centre, so they sit under the panels.
3. **Stand the sides** on the rails, feet over the rails, inside faces 350 mm apart. Fix each foot with two angle brackets at the engraved crosses.
4. **Braces:** from the engraved crosses on each crossbar down to the ends of the middle cross member, with an angle bracket each end.
5. **Bearings** on the outer faces. Drill the arm with the templates (axle 400 from the end, hang bolt 50 from the end), bolt on the doubler plates, then thread the axle: bearing, spacer, arm, spacer, bearing. Collars outside the bearings.
6. **Trigger:** glue the three trigger layers (slotted one in the middle), push the tabs into the trigger base, and screw the base to the back cross member under the cocked arm's trigger eye.
7. **Sensor plate:** strap to the arm 100–150 mm from the axle; strap the sensor to the plate.
8. Sling, pouch, beanbags and safety: Build Guide pages 5–9.

Also needed

Item	Qty
6 mm plywood, 900×600	4 sheets
Aluminium tube 40×20×1.6, 1400 long	1
6 mm aluminium plate, 120×40 (doublers)	4
2-bolt flange bearing, 20 mm bore + M10 bolts	2 + 4
20 mm steel axle, 420 long; 2 shaft collars	1
Spacer tube (25 mm conduit), 163 long	2
M12×50 hang bolt, nyloc nut, washers	1
M5×40 bolts for doublers	8
M6 eye bolts (sling eye, trigger eye)	2
10 mm trigger pin (rod or clevis pin)	1
Pine 90×45: rails 1500; cross members 800	2; 3
Pine 70×35 braces, about 560	2
Steel angle brackets 50 mm	8
6 mm dowel, 30 long	10
90 mm PVC pipe, halved, 1200 long (trough)	1
Rated strap cradle and shackle (Guide p. 9)	1

Before the first launch check: the arm swings a full turn by hand without touching anything; the bag clears the base by at least 80 mm at the bottom of its swing; the bearing bolts and the hang bolt are tight; and the frame is pegged or sandbagged. The aluminium doublers are **not** laser-cut — use the templates on sheet 4 to mark and drill them.